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ORIGINAL ARTICLE

Risk of Juvenile Idiopathic Arthritis Following HPV9 Vaccination: A Retrospective Cohort Study Before and During the COVID-19 Pandemic

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Abstract

Objective: To evaluate whether 9-valent human papillomavirus (HPV9) vaccination is associated with an increased risk of juvenile idiopathic arthritis (JIA), particularly during the coronavirus disease-2019 (COVID-19) pandemic.

Patients and Methods: We conducted a retrospective cohort study using TriNetX U.S. Collaborative Network data from January 1, 2016, to December 31, 2023. Girls aged 9-13 years who received their first HPV9 dose in either the prepandemic (2016-2019) or pandemic (2020-2023) period were matched with unvaccinated controls. Exclusion criteria included previous JIA diagnosis, antirheumatic drug use, or positive rheumatoid factor. Incidence of new-onset JIA was tracked over 8 days to 36 months. Cox regression and Kaplan-Meier survival analysis were used to evaluate hazard ratios (HRs) and JIA-free survival.

Results: Among 99,243 vaccinated and 1.1 million control individuals, HPV9 recipients had a significantly reduced risk of JIA at 36 months in both periods (HR 2016-2019, 0.207, $P<.001$; HR 2020-2023, 0.287, $P<.001$). No increased risk was observed during the early postvaccination period. The estimated cumulative probability of JIA did not differ significantly between vaccinated groups across the 2 periods ($P=.9$), nor among unvaccinated controls ($P=.238$), indicating no modifying effect from COVID-19.

Conclusion: The HPV9 vaccination was associated with a lower risk of JIA, and this effect can last at least for 3 years. The COVID-19 pandemic did not alter this relationship. These findings reinforce the immunological safety of HPV9 and provide reassurance for adolescent vaccination programs, even in pandemic contexts.

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Human papillomavirus (HPV) is a double-stranded DNA virus implicated in a wide range of pathologies, including cervical cancer, genital warts, and potentially autoimmune disorders.[1-3] Prophylactic vaccination, particularly the 9-valent HPV (HPV9) vaccine approved by the US Food and Drug Administration in 2014, has demonstrated high efficacy in preventing HPV-related diseases.[4,5] Although its clinical benefits are well established, concerns about vaccine-associated autoimmunity persist, especially in pediatric populations.[6,7]

Juvenile idiopathic arthritis (JIA) is the most common autoimmune rheumatic disease in children. It is characterized by chronic joint inflammation with a multifactorial etiology, involving both genetic predisposition and environmental triggers, including viral infections. Molecular mimicry between viral antigens and host tissues is a hypothesized mechanism linking infections to autoimmune disease onset.[8,9]

Several observational studies have explored a possible association between HPV vaccination and autoimmune diseases.[10-12] Analyses using the Vaccine Adverse Event

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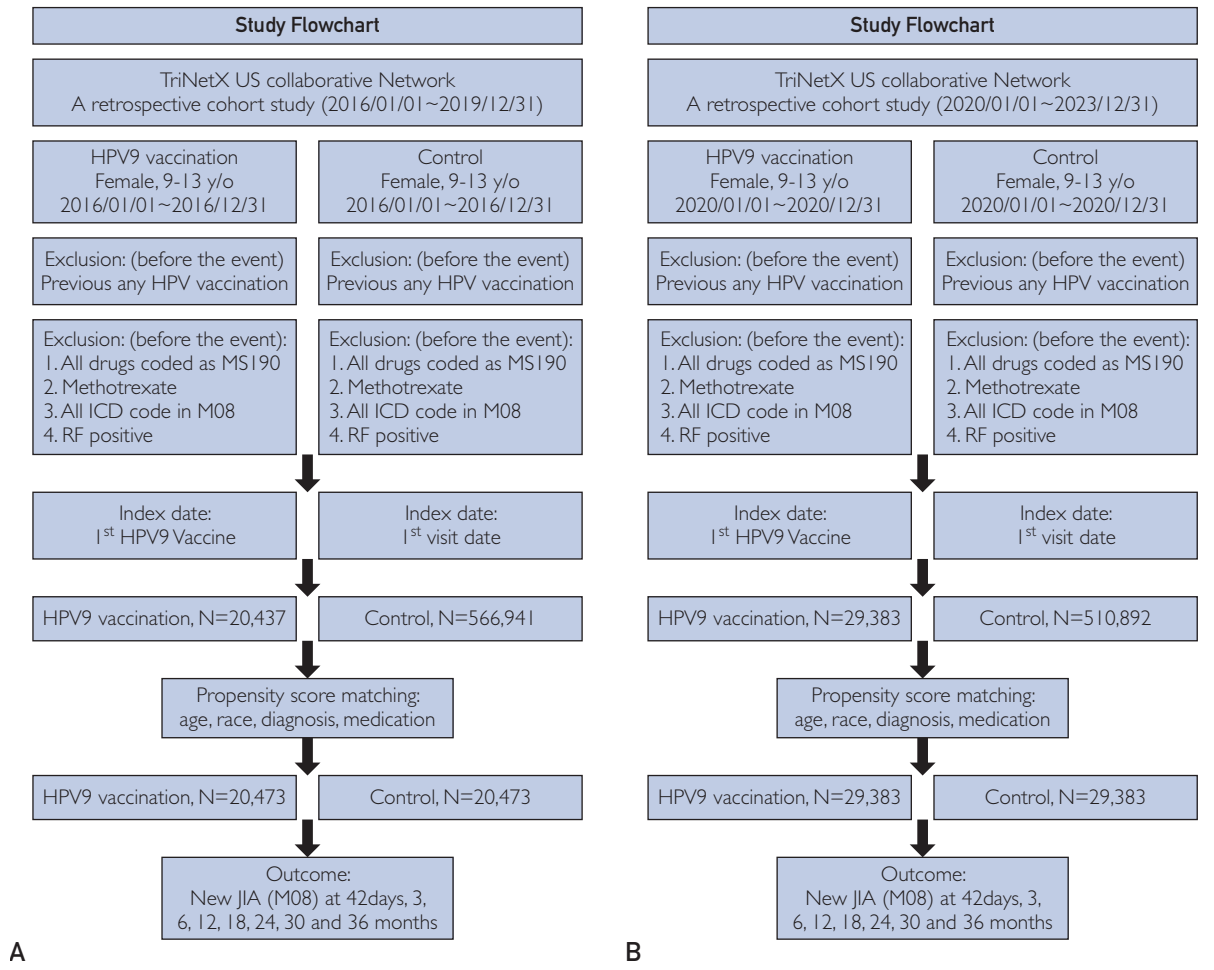


FIGURE 1. (A-D) shows study flowcharts for 4 cohort comparisons.

Reporting System (VAERS) and inverse probability weighting models in Colombia have reported increased risks of rheumatoid arthritis following HPV4 vaccination.[13,14] However, these studies were limited by self-reported outcomes and lacked robust clinical validation. In contrast, large-scale electronic health record-based studies using clinically diagnosed endpoints found no significant association between HPV4 and autoimmune conditions.[15]

To date, no published studies have specifically assessed the long-term safety of HPV9 vaccination with regard to JIA onset. Furthermore, the COVID-19 pandemic has disrupted health care access, vaccination schedules, and autoimmune disease surveillance, potentially influencing outcomes in ways yet to be characterized.[16]

To address this gap, we conducted a multicohort, retrospective study using the TriNetX U.S. Collaborative Network. We compared the incidence of JIA between vaccinated and unvaccinated adolescent females across prepandemic and pandemic periods. We aimed to determine whether HPV9 vaccination affects JIA risk and whether this association was influenced by the COVID-19 era.

PATIENTS AND METHODS

We used deidentified data from the TriNetX U.S. Collaborative Network, encompassing over 60 health care organizations and 20 million patients. Female participants aged 9-13 years old were included. Four analyses were performed:

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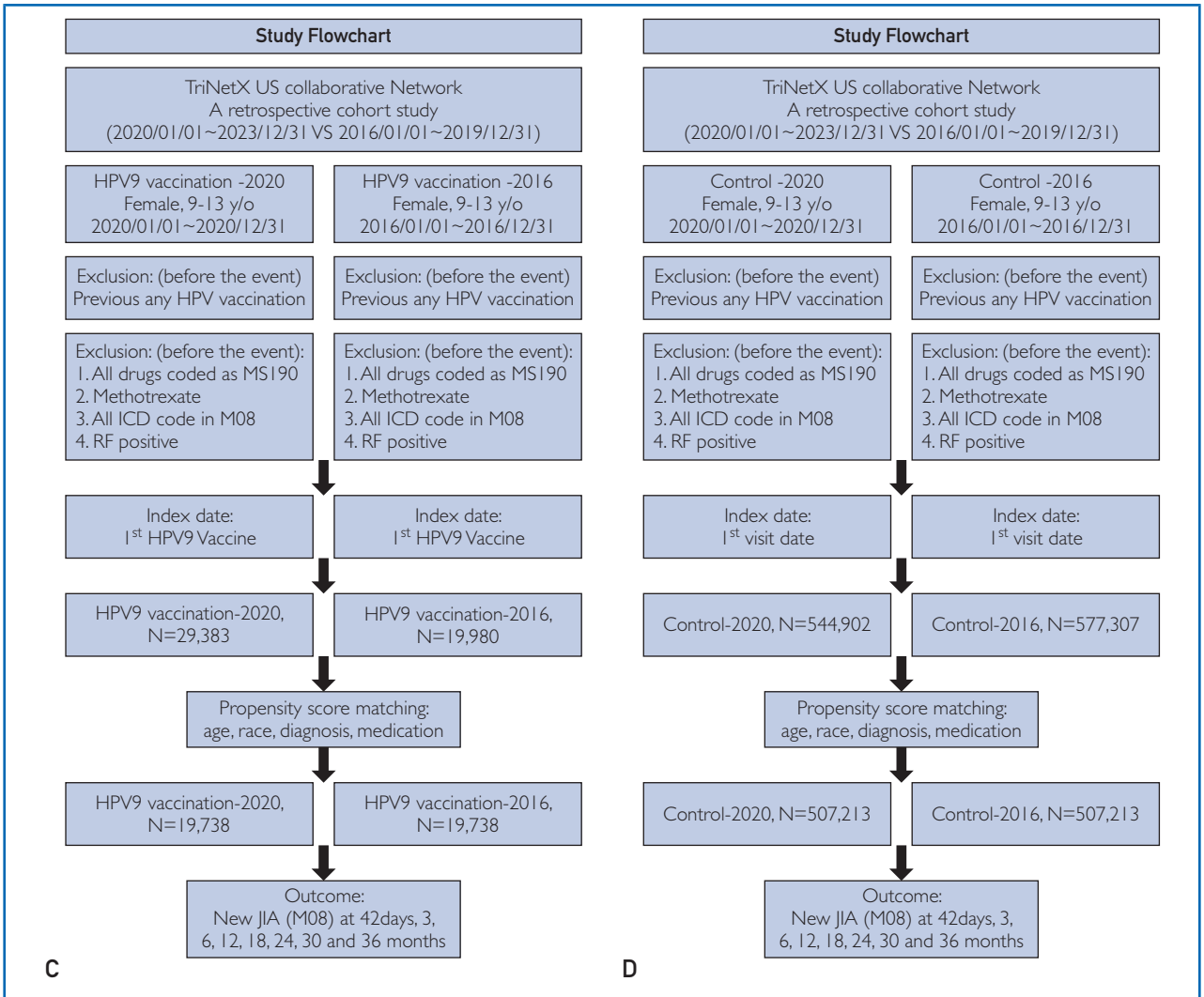


FIGURE 1. (*continued*).

1. HPV9-vaccinated versus unvaccinated (2016-2019)
2. HPV9-vaccinated vs. unvaccinated (2020-2023)
3. HPV9-vaccinated in 2016 versus 2020
4. Unvaccinated cohorts in 2016 versus 2020

The flow charts of these 4 analyses were shown as Figure 1A-D.

Exclusion criteria included previous JIA (International Classification of Diseases, 10th Revision, Clinical Modification M08.xx), positive rheumatoid factor, disease-modifying antirheumatic drugs for JIA including methotrexate, hyaluronate, leflunomide, adalimumab, etanercept, tocilizumab, golimumab, and updacinib. Propensity score matching was applied based on age, race, musculoskeletal diagnoses, and nonsteroidal anti-inflammatory drugs use.

All diagnosis, medication, procedure, and laboratory codes used for cohort definition and exclusion are listed in Supplemental Table 1, available online at http://www.mcpiqojournal.org.

Index events were defined as the first HPV9 vaccination (for vaccinated) or first clinical visit (for unvaccinated) within the study period. Follow-up intervals included 42 days, 3, 6, 12, 18, 24, 30, and 36 months. Events within 7 days of the index were excluded.

TABLE 1. Comparison of the Four Cohorts

Characteristic	HPV9 case (2020)	HPV9 control (2020)	HPV9 case (2016)	HPV9 control (2016)
Number of participants (n)	507,252	507,252	511,869	511,869
Age, mean (SDM)[a]	11.3 (1.3)	11.3 (1.3)	11.5 (1.3)	11.5 (1.3)
Female, n (%)	507,252 (100%)	507,252 (100%)	511,869 (100%)	511,869 (100%)
Race (White/B or AA[b]/Asian), (%)	50.3/19.4/4.8	56.8/14.8/3.5	52.4/22/3.4	57.7/16.2/3.2
Diagnosis (M00-M99),[c] n (%)	7,557 (25.7%)	107,545 (21.1%)	4,743 (23.2%)	79,073 (13.9%)
Medication (MS100),[d] n (%)	7,730 (26.3%)	107,545 (21.1%)	4,017 (19.7%)	79,984 (14.1%)

[a]SDM standardized mean difference.
[b]B or AA: black or African American.
[c]M00-M99: diseases of the musculoskeletal system and connective tissue.
[d]MS100: antirheumatics.

Cox proportional hazards models were used to estimate hazard ratios (HRs). Kaplan-Meier survival analysis and log-rank tests were performed to compare time-to-JIA diagnosis.

Ethics approval was granted by the Taichung Veterans General Hospital institutional review board (Approval number: CE23480C). Informed consent was waived due to anonymous data.

RESULTS

Baseline Characteristics

We established 4 cohorts (HPV9 and control groups in 2016-2019 and 2020-2023) based on the predefined criteria described in the Methods section. The baseline characteristics of these cohorts are summarized in Table 1. After applying 1:1 matching by age at initial event, race, diagnosis of musculoskeletal diseases (M00-M99), and nonsteroidal anti-inflammatory drugs use (MS100), we obtained 4 matched groups. These matched cohorts achieved exact balance across the matching variables (Supplemental Tables 2 and 3, available online at http://www.mcpiqojournal.org).

Cumulative probability and Risk of JIA

In both periods, vaccinated individuals exhibited a relatively lower risk of developing JIA when compared with controls. No cases of JIA were recorded in the first 6 months after vaccination. From 12 to 36 months, the JIA incidence increased in both groups but remained consistently lower in the HPV9 cohort (Figure 2A, B; Table 2).

- In the cohort from 2016 to 2019, estimated cumulative JIA probability in the vaccinated group was 0.043% at 36 months, compared with 0.201% in controls. The HR was 0.207 (95% CI, 0.096-0.447; $P<.001$).
- Kaplan-Meier analysis confirmed superior JIA-free survival in vaccinated individuals (Figure 2A).
- In the cohort from 2020 to 2023, estimated cumulative probability was 0.043% in the vaccinated group and 0.157% in controls at 36 months, with an HR of 0.287 (95% CI, 0.133-0.618; $P<.001$).
- The vaccinated group again showed significantly higher JIA-free survival (Figure 2B).

Detailed incidence rates, hazard ratios, and JIA-free survival data for the 2016 and 2020 cohorts are provided in Supplemental Tables 2 and 3.

Pandemic Period Comparisons

To assess the effect of the COVID-19 pandemic, 2 additional comparisons were made:

1. Vaccinated individuals from 2016 versus 2020 (n=19,738 each).
2. The estimated cumulative probability of JIA was 0.041% in the 2020 group and 0.044% in the 2016 group, with no significant difference (HR=0.937; 95% CI, 0.340-2.585; $P=.9$).
3. The Kaplan-Meier curve for this comparison is shown in Figure 2C.
4. Unvaccinated controls from 2016 versus 2020 (n=507,213 each).

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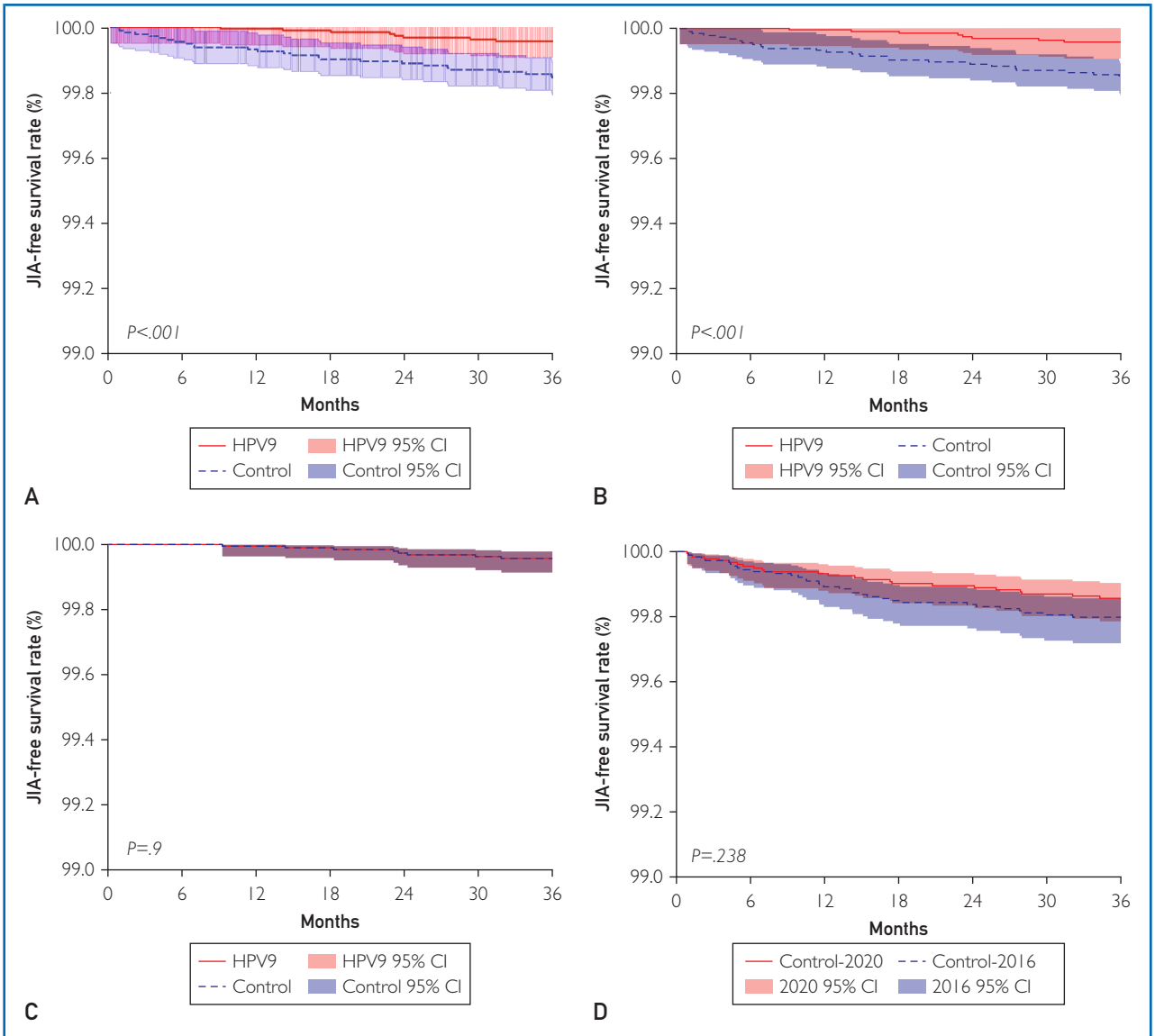


FIGURE 2. (A-D) Kaplan-Meier curves of JIA-free survival stratified by vaccination status and time. JIA, juvenile idiopathic arthritis.

5. The estimated cumulative probability of JIA at 36 months was 0.150% in 2016 and 0.141% in 2020, with an HR of 0.930 (95% CI, 0.824-1.049, *P*=.238), indicating no significant change.

Corresponding survival curves are illustrated in Figure 2D.

Survival curves and statistical comparisons for inter-period cohort analyses are presented in Supplemental Tables 4 and 5, available online at http://www.mcpiqojournal.org. These results suggest that the COVID-19 pandemic did not significantly impact the observed association between HPV9 vaccination and JIA risk.

DISCUSSION

In this large-scale, retrospective cohort study using the TriNetX U.S. Collaborative Network, we found that HPV9 vaccination was associated with a significantly lower risk of developing JIA between 12 and 36 months postvaccination. This finding was consistent

TABLE 2. HRs and 95% CIs for the Risk of Juvenile Idiopathic Arthritis (JIA)

	Cumulative probability(%) of JIA since index date									
	Patients with outcome (at 36 mo)	42 d	3 mo	6 mo	12 mo	18 mo	24 mo	30 mo	36 mo	HR (95% CI)
Period 1: 2016 to 2019										
HPV9 (n=20,437)	10	0.000	0.000	0.000	0.005	0.010	0.032	0.038	0.043	0.207 (0.096-0.447)
Control (n=20,437)	34	0.016	0.028	0.056	0.109	0.157	0.169	0.195	0.201	Reference
Period 2: 2020 to 2023										
HPV9 (n=29,383)	10	0.000	0.000	0.000	0.005	0.010	0.032	0.038	0.043	0.287 (0.133-0.618)
Control (n=29,383)	24	0.017	0.022	0.045	0.068	0.098	0.111	0.130	0.157	Reference
HPV9 cohorts in different periods										
HPV9-2020 (n=19,738)	10	0.000	0.005	0.005	0.005	0.017	0.034	0.034	0.041	0.937 (0.340-2.585)
HPV9-2016 (n=19,738)	10	0.000	0.000	0.000	0.005	0.011	0.033	0.039	0.044	Reference
Control cohorts in different periods										
Control-2020 (n=507,213)	464	0.014	0.024	0.041	0.068	0.088	0.105	0.120	0.141	0.930 (0.824-1.049)
Control-2016 (n=507,213)	608	0.013	0.023	0.044	0.069	0.092	0.113	0.131	0.150	Reference

Abbreviation: HR, hazard ratio.

across both the prepandemic and pandemic periods. Notably, no increased risk was observed during the early postvaccination period (42 days, 3 months, and 6 months), and Kaplan-Meier analyses confirmed a significantly higher JIA-free survival in vaccinated individuals compared with matched controls.

To our knowledge, this is the first study to assess the incidence of JIA after HPV9 vaccination. Our study design incorporated a carefully chosen early risk window (8-42 days), based on immunopathological considerations and regulatory reporting guidelines.[17] Autoimmune reactions typically require more than 1 week for manifestation after an immunologic stimulus. Furthermore, 42 days is the standard period used by the US VAERS for arthritis surveillance after live viral vaccinations.[18] This timeframe balances sensitivity to true vaccine-related signals while minimizing unrelated temporal associations.

Epidemiological studies have reported varied JIA incidence rates worldwide. The US registry data from 1996 to 2009 estimated an incidence of 16.4 per 100,000 girls, whereas pooled estimates across 34 studies reported standardized rates ranging from 3.6 to 23 per 100,000.[19,20] Our control groups showed notably lower rates, with JIA incidence declining from 5.5 per 100,000 at 12 months to 1.8-2.0 per 100,000 at 36 months. These differences may reflect changes in diagnostic practices, population selection, or evolving vaccine coverage. Importantly, our study population focused exclusively on adolescent girls aged 9-13 years, the primary target group for HPV vaccination.

The immunological link between HPV and autoimmune conditions has been increasingly explored. The HPV infection has been associated with rheumatoid arthritis (RA),[19,21] and high rates of autoantibodies have been detected in HPV DNA-positive patients.[22] Mechanistic hypotheses include molecular mimicry, with HPV peptides shown to elicit cross-reactive immune responses to self-antigens.[23] Notably, citrullinated peptides derived from HPV-47 have been found to stimulate autoantibodies in patients with RA.[24] Conversely, individuals with autoimmune diseases may be more susceptible to persistent HPV infection due to immune dysregulation.[25]

Vaccination against HPV has been reported to possess durable protective effects against infection and the downstream risk of uterus cervical cancer; and may possibly reduce autoimmune activation.[26,27] The National Health and Nutrition Examination Survey data suggest a reduced incidence of RA in vaccinated adults, although that study relied on self-reported outcomes.[28] Our findings extend these observations to a pediatric population, indicating that HPV9 vaccination may mitigate immune activation pathways that contribute to JIA pathogenesis.

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Our study also addressed the potential modifying effect of the COVID-19 pandemic. Despite widespread changes in healthcare access and infection exposures, we observed no difference in JIA incidence between 2016 and 2020 among either vaccinated or unvaccinated cohorts. This suggests that the protective effect of HPV9 against JIA was robust and not confounded by pandemic-related factors. Previous studies in adult populations have reported mixed trends in autoimmune disease incidence during COVID-19.[29,30] However, pediatric data remain limited, and our results provide important insights into disease stability during this period.

Limitations

Several limitations should be acknowledged. First, while TriNetX provides access to a large and diverse population, the dataset is US-based, and its racial composition (White: 50%-58%, Black: 14%-22%, and Asian: 3%-5%) may not represent global populations. Second, although cohorts were matched for age and comorbidities, index event dates could not be fully synchronized. This may have introduced seasonal effects that could impact short-term incidence assessments, although long-term trends were likely unaffected. Third, JIA diagnoses were based on ICD-10 coding with or without antirheumatic prescriptions; while this enhances specificity, misclassification remains possible. Finally, data on family history, HLA genotypes, or environmental exposures were not available, limiting adjustment for genetic or lifestyle risk factors.

CONCLUSION

In this large retrospective study, HPV9 vaccination was not associated with an increased risk of JIA. A lower incidence of JIA was observed in the vaccinated groups, which may help reassure the public regarding the immunological safety of HPV9 in adolescent girls. These findings remained consistent across both prepandemic and pandemic periods, suggesting the robustness of the observed association. Although the results are encouraging, they do not establish a causal protective effect, and further prospective studies are needed to clarify the potential immunomodulatory role of HPV vaccination.

POTENTIAL COMPETING INTERESTS

The authors report no competing interests.

ETHICS STATEMENT

This study was approved by the Institutional Review Board of Taichung Veterans General Hospital (Approval No. CE23480C). Informed consent was waived due to use of de-identified data.

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SUPPLEMENTAL ONLINE MATERIAL

Supplemental material can be found online at http://www.mcpiqojournal.org. Supplemental material attached to journal articles has not been edited, and the authors take responsibility for the accuracy of all data.

Abbreviations and Acronyms: **COVID-19,** coronavirus disease 2019; **HPV,** human papillomavirus; **HPV9,** 9-valent human papillomavirus vaccine; **HR,** hazard ratio; **JIA,** juvenile idiopathic arthritis; **RA,** rheumatoid arthritis; **VAERS,** Vaccine Adverse Event Reporting System

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